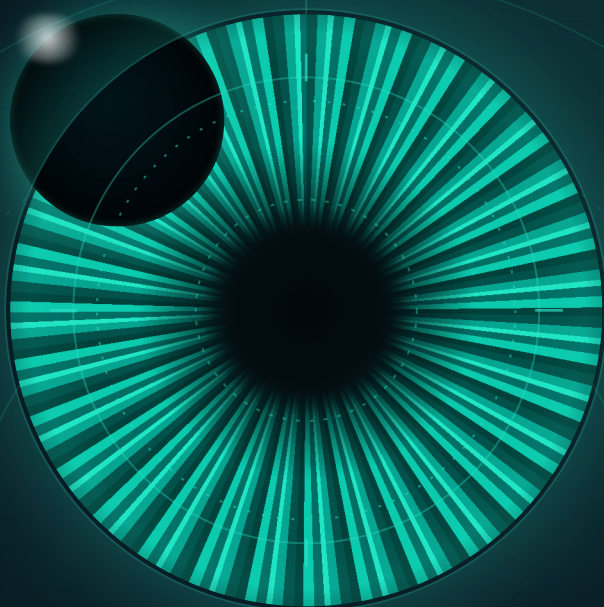




EyeBot

*An AI training companion for the whole of ophthalmic education —
grounding every learner, from first-year student to practising
clinician, in faithful, adaptive, measurable practice.*



PREPARED FOR THE 2026 INTERNATIONAL E-LEARNING AWARDS · ACADEMIC DIVISION
· E-LEARNING EXPERIENCES

01 EXECUTIVE SUMMARY

Imagine every ophthalmic trainee — whatever their role — with a tutor who never sleeps, a patient who is always available, and a curriculum that adapts to exactly what they keep forgetting.

SNEC EyeBot is an AI learning companion that transforms a clinical curriculum into three evidence-based experiences — a Socratic tutor, inexhaustible virtual-patient simulations, and a self-optimising memory engine — woven into one adaptive loop. Grounded today in SNEC's clinical curriculum and architected to serve the **full spectrum of eye-care learners**, it also gives educators a live, cohort-wide view of mastery for the first time. The platform is built, demonstrable, and ready to pilot.

3-in-1



Tutor, clinical simulator and memory engine working as a single adaptive loop.

Grounded



Every answer drawn from validated clinical material — faithful, not the open internet.

Cohort



Live at-risk detection, retention scoring and weekly digests for educators.

A fixed curriculum becomes a **living, personalised, measurable** learning experience — and positions SNEC at the front of AI-enabled clinical education.

02 THE CHALLENGE

Training the eyes behind the eye care.

Across the eye-care continuum — medical students on rotation, optometry students, ophthalmology residents, nurses and allied-health technicians — training is rigorous, yet it meets four structural limits that no amount of dedication can fully solve.

**Expert time is scarce**

One educator cannot give every trainee unlimited one-to-one tutoring or on-demand case practice.

**Case exposure is uneven**

Which conditions a trainee actually meets is left to the luck of the clinical roster.

**Knowledge decays**

Without deliberate reinforcement, much of what is taught is forgotten within weeks of the lesson.

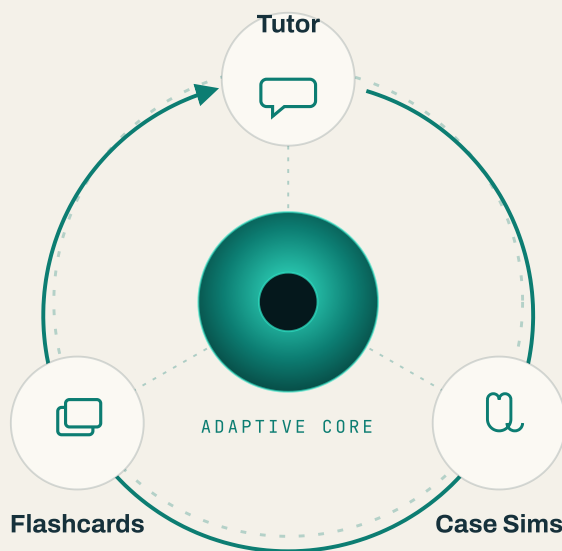
**Cohort gaps are invisible**

Educators often discover weaknesses at assessment time — when it is already late to intervene.

EyeBot is engineered to dissolve all four — **at once.**

03 THE SOLUTION

One companion. Three modes. A single adaptive loop.



MODE 01

Socratic Tutor

Plain-language questions met with Explanation → Mechanism → Clinical Pearl → a check for understanding, always grounded in the clinical knowledge base.



MODE 02

Case Simulations

Interview a virtual patient, order investigations, commit to a diagnosis and plan — graded on a four-domain rubric out of 40 with instant feedback.



MODE 03

Spaced Flashcards

High-yield cards auto-generated from each session and every missed clinical point, scheduled by the SM-2 algorithm for durable recall.

...and quietly, an analytics layer watches the whole cohort — turning individual practice into institutional insight.

04 THE INNOVATION

An innovation — not an integration.

EyeBot is not a chatbot bolted onto a syllabus. Five deliberate advances make it genuinely new for clinical education.

01



Curriculum-grounded clinical AI

Answers are retrieved from the institution's own validated material — faithful and auditable, never the untraceable guesswork of a generic model.

02



Inexhaustible virtual patients

The AI plays both patient and examiner: unlimited, standardised case exposure with rubric-graded feedback — independent of roster luck.

03



A self-optimising loop

Each learner's weak points automatically become their next flashcards and tougher cases — genuine personalisation, at scale.

04



Reliable by architecture

A Workflows–Agents–Tools design separates AI judgement from deterministic execution — trustworthy enough for clinical training.

05



Cohort intelligence

Learning analytics purpose-built for clinical education: at-risk detection, retention scoring and learning-velocity — live, not at term's end.

Together: the first AI companion built for **how**
eye care is actually taught.

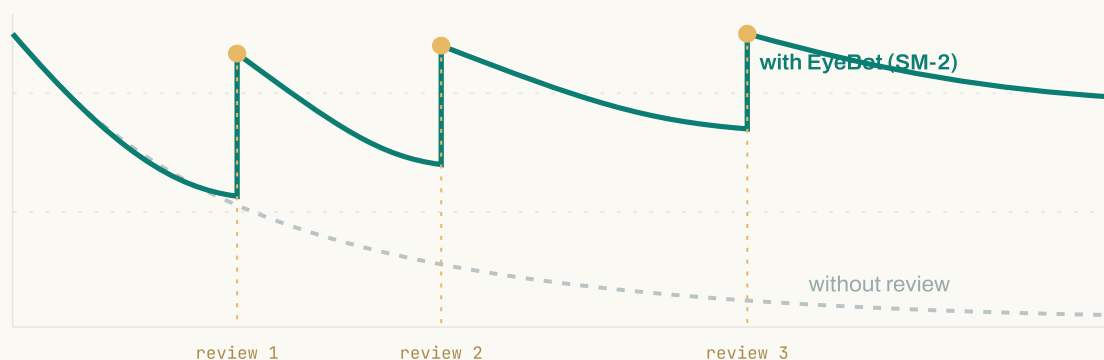
05 PEDAGOGICAL FOUNDATION

Grounded in learning science, not novelty.

Every design decision in EyeBot maps to an established principle of how clinicians actually learn and retain.

FIG. 1 — SPACED REPETITION VS. THE FORGETTING CURVE

Retention ↑ • Time →

**Retrieval & spacing**

SM-2 spaced repetition fights the forgetting curve — each review lengthens durable retention.

**Case-based reasoning**

Simulations rehearse the full clinical chain: history, investigation, diagnosis, management.

**Socratic scaffolding**

The tutor explains mechanism, then checks understanding — active recall over passive reading.

**Formative feedback**

Rubric scores and targeted cards turn every mistake into the learner's next rep.

One engine. Every learner.

EyeBot adapts its content, cases and rubric to each role — and the same engine extends across specialties and institutions. What begins with SNEC's programme scales to the whole of ophthalmic education.



Role-aware

Content, case libraries and rubrics tuned to each learner's scope of practice.



Specialty-extensible

The same engine accepts new curricula — glaucoma, retina, paediatrics and beyond.



Institution-ready

PDPA-compliant consent, bcrypt credentials and role-based access — privacy by design.

From a single programme to an **institution-wide standard** for how eye-care professionals are trained.

From prototype to proof.

PROPOSED SNEC PILOT

COHORT

A representative intake of ophthalmic trainees over an 8–12 week module, benchmarked against a prior cohort as historical control.

METRICS

Pre/post knowledge test · case-rubric progression · flashcard retention · engagement (active days & streaks) · educator time saved · learner satisfaction.

METHOD

Baseline → guided daily use → measured uplift, with the analytics layer capturing evidence automatically throughout.

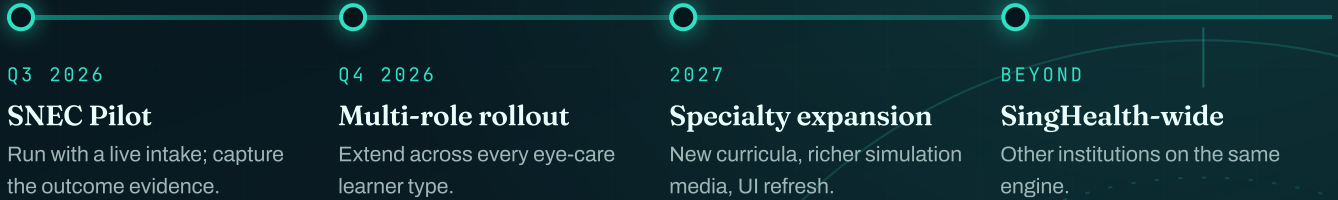
FIT TO IELA CRITERIA

ATTRIBUTE	HOW EYEBOT DELIVERS
Educational soundness PEDAGOGY	Curriculum-grounded; SM-2 spacing; Socratic + case-based design rooted in learning science.
Effectiveness OUTCOMES	Closes gaps in a measured loop; analytics quantify uplift at learner and cohort level.
User experience ENGAGEMENT	Polished, gamified and mobile-friendly, with daily check-ins that build habit.
Significance IMPACT	First AI companion spanning the ophthalmic learning continuum; scalable across SingHealth.

Because impact is **measured, not asserted**, the pilot doubles as the evidence base for the award submission itself.

08 ROADMAP & VISION

Where this goes next.



SNEC EyeBot turns a fixed curriculum into a **living, personalised and measurable** learning experience — and positions SNEC at the forefront of AI-enabled clinical education.

ENGINEERED WITH

FastAPI

Supabase

React + TypeScript

Anthropic Claude

Workflows · Agents · Tools